

GreenLight Laser (Photoselective Vaporization, PVP)

Patient Information · A laser surgery for an enlarged prostate, with little bleeding

WARWIKI

GreenLight laser surgery treats an enlarged prostate (BPH) that blocks the flow of urine. The surgeon works through a scope passed up the urethra, so there is no cut on the skin. A laser turns the blocking tissue into vapor and seals blood vessels as it goes. Most men get a stronger stream, with **less bleeding** and a **shorter catheter time** than with TURP.

About This Procedure

The green laser light is absorbed by blood in the prostate tissue. This heats the tissue so it vaporizes, layer by layer, and opens a wide channel for urine. The outer part of the prostate stays in place. In a large trial, results at 2 years were as good as TURP, the long-standing standard operation.

Benefits and Risks in Numbers

6 in 10

had the catheter out the day after surgery (expert centers)

8 in 10

needed no further prostate treatment 10 years later (9 in 10 at 5 years)

about 7 in 10

had little or no semen afterward in a small trial (harmless)

Numbers are averages from published studies. Your own risk may differ; ask your team.

Is It Right for Me?

- The best evidence is for **small to medium prostates**. Very large prostates more often need another treatment later; HoLEP or simple prostatectomy may suit them better.
- Because it bleeds little, it can be an option if you take **blood thinners**. Your team will tell you whether to stop them.
- **No tissue is saved** for the lab, so it is not checked for prostate cancer.
- Burning and urgency in the first weeks are more common than after TURP.
- Over 5 to 10 years, more men need a repeat procedure after GreenLight than after HoLEP.

Learn the Terms

BPH

An enlarged prostate that is not cancer.

PVP

Photoselective vaporization of the prostate. Another name for GreenLight.

Vaporize

To turn tissue into vapor with heat.

Retrograde ejaculation

Semen goes back into the bladder instead of out. It is common and harmless.

Catheter

A soft tube that drains the bladder for a day or two after surgery.

TURP and HoLEP

Other operations for BPH that remove prostate tissue through a scope.

What changes afterward? Most men get a much stronger stream. The most common lasting change is retrograde ejaculation: little or no fluid comes out. You usually still have an orgasm. It is harmless, but it can make it harder to father a child. A technique that spares the area around the ejaculation ducts may lower this risk; ask your surgeon. In studies, erection scores were similar to TURP over the first year and slightly lower at 2 years.

Same day

Many men go home the same day or the next morning, with a catheter.

Days 1–3

The catheter comes out, often after 1–2 days.

First few weeks

Burning, urgency and some blood in the urine. Drink fluids.

Weeks 2–3

A small bleed can happen as scabs come off. Rest and drink more.

How to Get Ready

- Follow your instructions about eating, drinking and which medicines to stop, including blood thinners. Do not stop them on your own.
- A **urine test** checks for infection first.
- Arrange a ride home.

What Happens

- 1 You get general or spinal anesthesia, so you feel nothing.
- 2 A scope is passed up the urethra, and a thin laser fiber goes through it.
- 3 The laser vaporizes the blocking tissue, sealing blood vessels as it works.
- 4 A catheter is placed to drain the bladder while it settles.

After Surgery

- Passing urine may sting and you may go often for the first days after the catheter comes out. This improves.
- Avoid straining and constipation while it heals.
- If your stream gets weak again months or years later, tell your team. Scar tissue or regrowth can be treated.

Call the office that did your surgery if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Bleeding or burning that gets worse instead of better

Go to an emergency department if:

- You cannot urinate for several hours and your bladder feels full or painful
- Your urine looks like red wine or has clots, or the catheter stops draining
- You cannot reach the office and are worried

Three Things to Remember

1. GreenLight vaporizes the blocking prostate tissue through a scope, with little bleeding.
2. The catheter usually comes out in a day or two. Burning and urgency settle over a few weeks.
3. Little or no semen afterward is common and harmless.